Graph Attention Networks (GAT)

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Introduction

GATs:

- Graph Attention Networks performs attention over local neighbors, unlike LLMs which compute attention over all tokens.

Project Goal:

- To reproduce the results of all transductive and inductive learning tasks tested by the original paper using GATs

Motivation for GATs:

- Use GATs to outperform MLP and CNN (GCN) across several graph learning tasks

Demonstrating Graph Learning Tasks

Transductive Learning

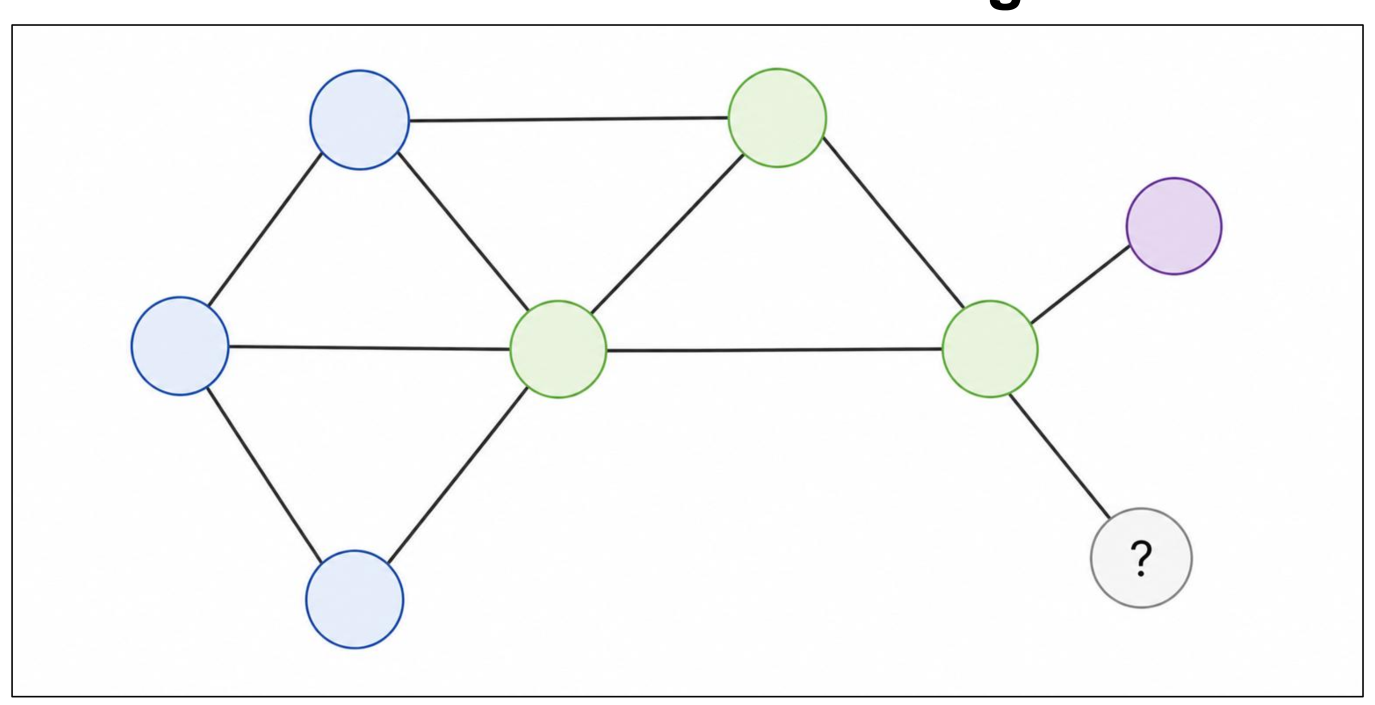

Figure 1: Demonstration of transductive learning

Inductive Learning

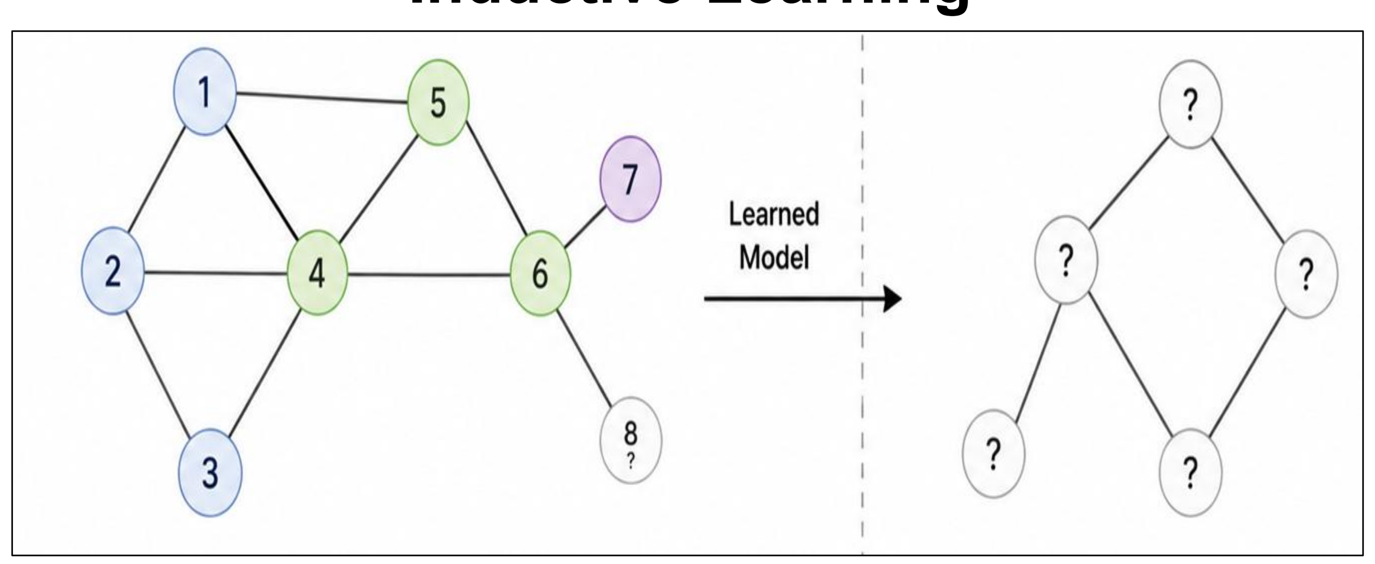

Figure 2: Demonstration of inductive learning

Methodology

Overview:

- We created a Jupyter Notebook to run the reimplementation using Google Colab with the following architecture:

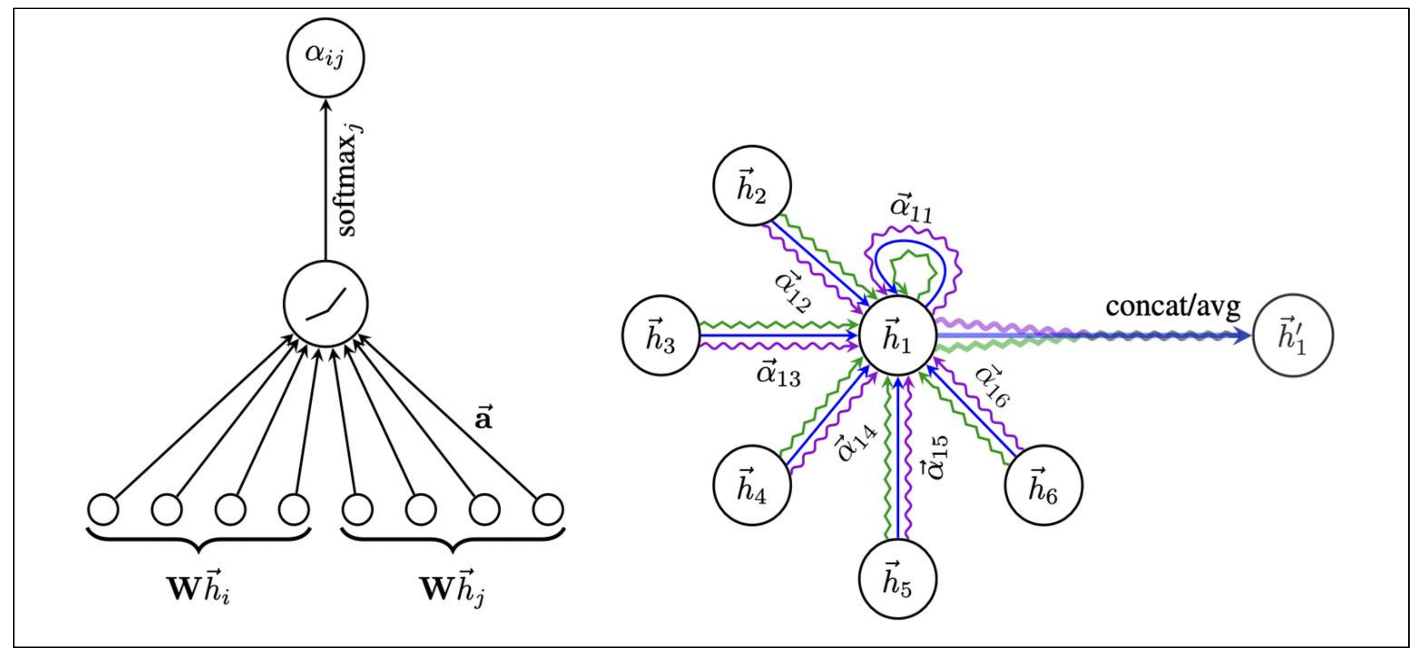

Figure 3: Representation of a GAT layer

	Cora	Citeseer	Pubmed	PPI
Task	Transductive	Transductive	Transductive	Inductive
# Nodes	2708 (1 graph)	3327 (1 graph)	19717 (1 graph)	56944 (24 graphs)
# Edges	5429	4732	44338	818716
# Features/Node	1433	3703	500	50
# Classes	7	6	3	121 (multilabel)
# Training Nodes	140	120	60	44906 (20 graphs)
# Validation Nodes	500	500	500	6514 (2 graphs)
# Test Nodes	1000	1000	1000	5524 (2 graphs)

Table 1: Summary of the datasets used

Modifications:

- Stricter early stopping
- Less maximum epochs
- Manual garbage collection to avoid out-of-memory errors

References

Veličković, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., & Bengio, Y. (2017). Graph attention networks. *arXiv preprint arXiv:1710.10903*.
Figures 1 and 2 are generated by ChatGPT
Figure 3 is from the original paper
Figures 4 and 5 are from the Jupyter Notebook from our GitHub repository
Table 1 is from the original paper
Tables 2 and 3 cite results from the original paper and our Jupyter Notebook

Results

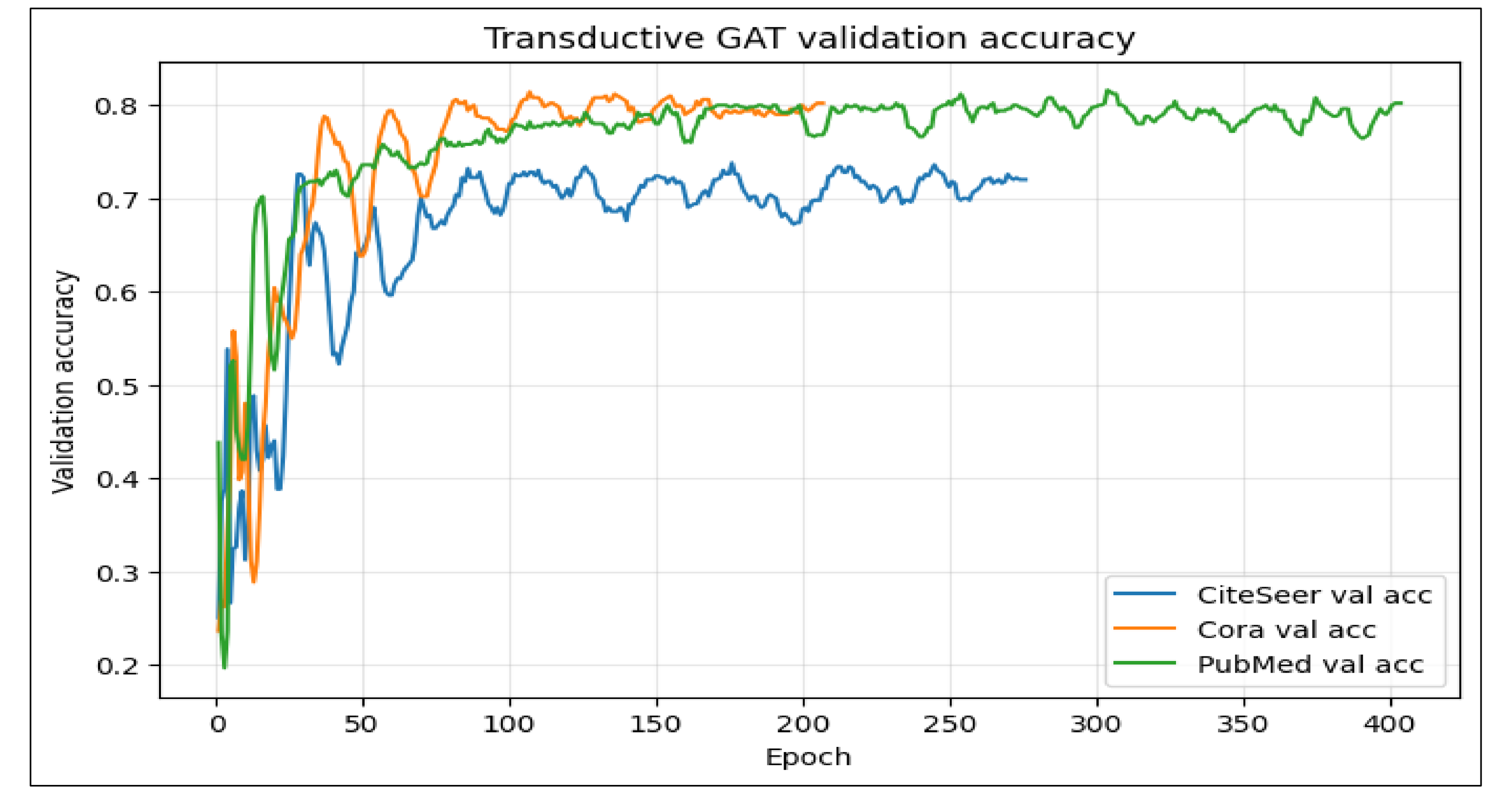

Figure 4: Transductive validation accuracy

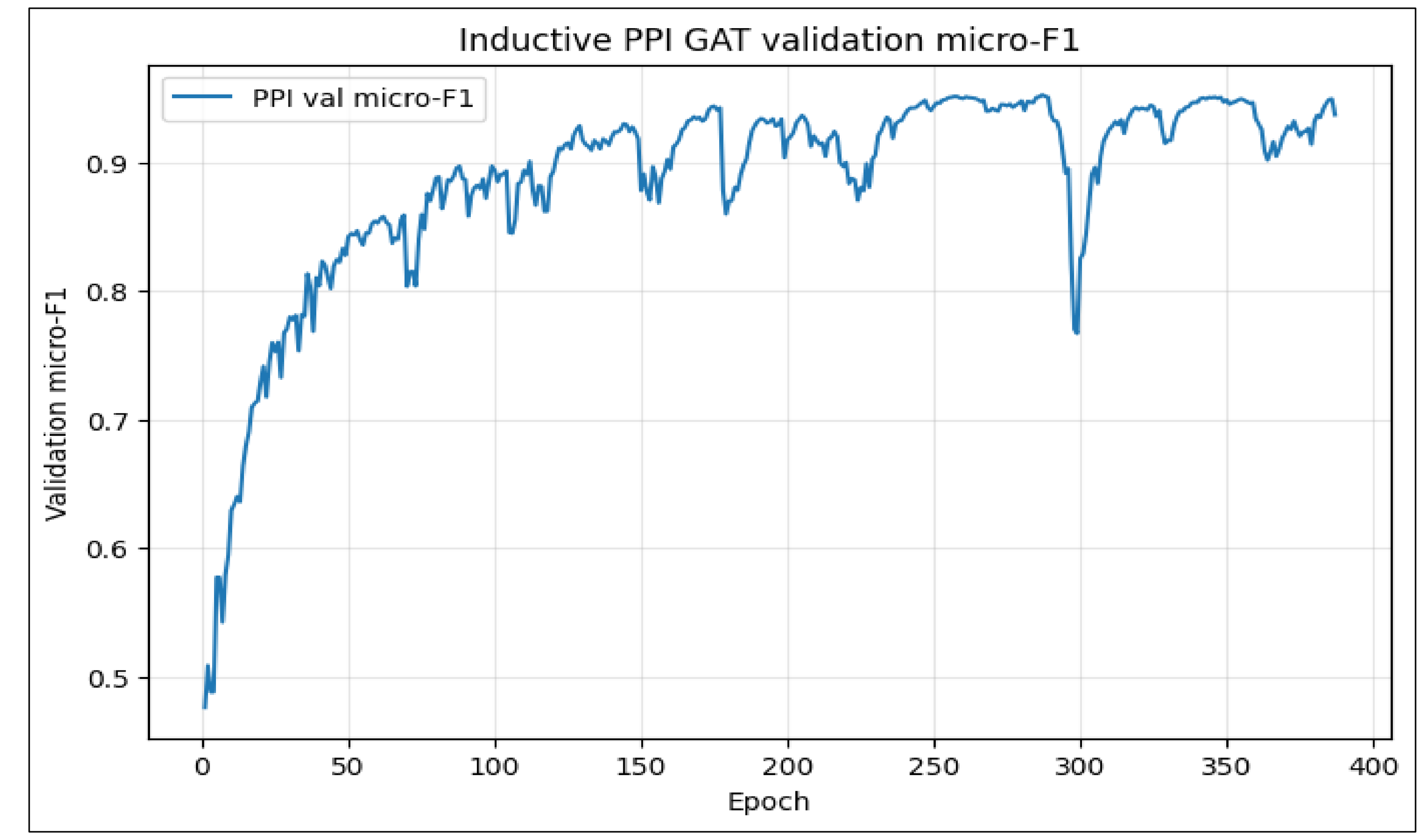

Figure 5: Inductive validation accuracy

Transductive			
Method	**Cora**	**Citeseer**	**Pubmed**
MLP	55.10%	46.50%	71.40%
GCN-64	81.4±0.5%	70.9±0.5%	79.0±0.3%
GAT (paper)	83.0±0.7%	72.5±0.7%	79.0±0.3%
GAT (ours)	**82.00%**	**71.80%**	**77.70%**

Table 2: Transductive test accuracies

Inductive	
Method	**PPI**
MLP	0.422
Const-GAT	0.934±0.006
GAT (paper)	0.973±0.002
GAT (ours)	**0.97**

Table 3: Inductive test accuracy

Conclusion

- Our implementation is reasonably close to the original results and outperformed the baselines in all learning tasks.
- The gaps in our results and the original results are likely due to the fact that our results represent a single test on one seed while the paper represents an average of multiple runs and we used less training epochs

Future Work

- Try out several different GAT model structures to outperform the paper's results
- Experiment with more random seeds and average the results